

Chittagong University of Engineering and Technology
Department of Computer Science and Engineering
B. Sc. Engineering Level-3, Term-I, Exam. 2024

Course No.: CSE-335
 Course Title: Operating Systems
 Marks: 210
 Time: 3 Hours

The figure in the right margin indicates full marks. The questions are of equal value. Answer any three questions from each section. Use separate script for each section.

Section-A

- Q.1(a) Create a diagram illustrating different process states and transitions. Use your diagram to showcase the sequence of states a process might go through. 10
- Q.1(b) Explained Tree-Structured, Acyclic-Graph, and General Graph Directories Structure with necessary diagrams. 15
- Q.1(c) Compare and contrast different IPC mechanisms. 10
- Q.2(a) What criteria are used to measure the performance of a CPU scheduling algorithm? 10
- Q.2(b) Explain the effect of increasing and decreasing the time quantum for the round robin scheduling algorithm. 10
- Q.2(c) Consider the following set of processes with the length of the CPU-burst time given in milliseconds: 15

Process	Arrival Time	Burst Time	Priority
P ₁	0	7	3
P ₂	2	4	3
P ₃	4	1	4 (Lowest)
P ₄	5	4	1 (Highest)
P ₅	3	2	1 (Highest)

- (i) Draw the Gantt Charts illustrating the execution of these processes using Priority, Preemptive and SRTF Scheduling algorithm.
- (ii) Compute the average turnaround time, waiting time, and response time for the two scheduling algorithm. 2.6, 7.1, 7.6
- Q.3(a) Given a resource allocation graph with processes P₁, P₂, P₃, and resources R₁, R₂, R₃. Draw a scenario that leads to deadlock. Explain why deadlock occurs and how it can be detected. 10
- Q.3(b) Consider a system with twelve magnetic tape drives and three processes P₁, P₂, and P₃. Process P₁ requires maximum ten tape drives, process P₂ may need as many as four tape drives and P₃ may need up to nine tape drives. Suppose that at time t₁, process P₁ is holding five tape drives, P₂ is holding two tape drives and P₃ is holding three tape drives. At time t₁, system is in: (Justify your answer) 10
- (i) safe state
- (ii) unsafe state
- (iii) deadlocked state
- (iv) starvation condition
- Q.3(c) Answer the following questions: 15
- (i) Consider the following Resource Allocation Graph (RAG):

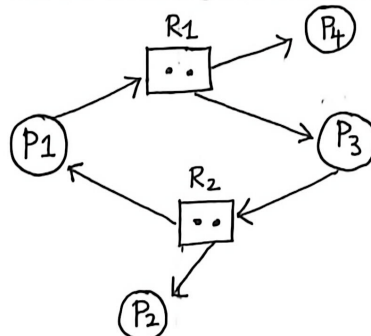


Fig. 3(c) (i)

If P_2 makes an additional request for R_1 , can the system grant it safely?

- (i) A cycle in the RAG is considered a necessary condition for deadlock. Is it always a sufficient condition? Briefly explain your answer.

Q.4(a) Answer the following questions:

15

- (i) Fill in the following table for RAID 0, RAID 1, RAID 4, and RAID 5.

RAID Level	Capacity	Reliability
RAID 0		
RAID 1		
RAID 4		
RAID 5		

- (ii) How does RAID 4 differ from RAID 5 in terms of parity placement?
 (iii) In the best case, how many disk failures can RAID 1 tolerate without data loss, and why?

Q.4(b) A disk drive has 3000 cylinders numbered from 0 to 2999. The queue of pending requests, in FIFO order is

20

450, 200, 900, 100, 500, and 600.

Assume that the initial position of the R/W head is at cylinder 550. Now, find the total distance the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms:

- (i) ~~FIFO~~ FIFO
 (ii) C-LOOK
 (iii) SSTF
 (iv) C-SCAN
 (v) LOOK.

Section B

Q.5(a) Answer the following questions:

15

- (i) What is the purpose of memory management unit (MMU) in main memory?
 (ii) Discuss the differences between external and internal fragmentation?
 (iii) How does TLB affect effective memory access time?

Q.5(b) With a neat sketch, explain the structure and working of an 'Inverted Page Table' in memory management.

10

Q.5(c) Consider the following segment table with each segment's base address and limit.

10

Segment	Base Address	Limit
0	520	500
1	350	150
2	100	200
3	1800	600
4	1050	650

Table 5 (c): Segment Table

Now, calculate the physical address for the following logical address:

- (i) 2, 50
 (ii) 4, 700
 (iii) 0, 400
 (iv) 1, 100
 (v) 3, 500

Q.6(a) Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs.

10

Q.6(b) How many page faults occur for the LRU page replacement algorithm for the following reference string with four page frames?

10

1, 2, 1, 2, 3, 4, 5, 3, 4, 1, 6, 7, 8, 7, 9, 7, 8, 9, 5, 1.

Q.6(c) Write short notes on:

15

- (i) Copy-on-write
 (ii) Thrashing
 (iii) Global vs. Local Allocation
 (iv) Belady's Anomaly
 (v) Demand Paging

- Q.7(a) Consider the paging example of the following figure for a 32-byte memory with 4-byte pages. 10

0	a
1	b
2	c
3	d
4	c
5	f
6	g
7	h
8	i
9	j
10	k
11	l
12	m
13	n
14	o
15	p

Logical memory

0	5
1	6
2	1
3	2

page table

Fig. 7(a)

Now map logical addresses 13, 3, 8, 6, and 15 to physical addresses. (Show your calculations)

- Q.7(b) Answer the following questions: 15

- Define compactions and explain why it is used?
- Compare the memory organization schemes of contiguous memory allocations and paging with respect to the following issues:
External Fragmentation
Internal Fragmentation

- Q.7(c) A system has the following parameters: 10

Memory access time: 100 ns

Page fault service time: 8 ms

Page fault rate: 0.1%

Calculate the Effective Memory Access Time (EMAT) for this demand paging system.

- Q.8(a) Consider P_1 , P_2 , and P_3 processes that require S_1 to happen before S_2 and S_2 to happen before S_3 . Now ensure synchronization among different processes using semaphore. 15

P_1 :	P_2 :	P_3 :
:	:	:
S_1	S_2	S_3
:	:	:

Figure 8(a)

- Q.8(b) Briefly explain the many-to-many, one-to-one, and many-to-one multithreading modules. 10

- Q.8(c) Briefly explain the following topics: 10

- Micro kernel approach of an OS design
- Concurrency vs parallelism
- Dual-mode operation in OS.

-The End-

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SECTION-A

- Q.1(a) Describe the actions taken by a kernel to context-switch between processes. 15
- Q.1(b) Let us consider five processes: P₁, P₂, P₃, P₄, and P₅ that take 16ms, 13ms, 10ms, 13ms, and 11ms to execute, respectively. The arrival time for the processes is 0, 6, 10, 8, and 33 milliseconds. P₁ has higher priority than P₂, P₄ and P₅. P₃ has a higher priority compared to P₁. P₄ and P₅ have equal priority. P₂ has the lowest priority; if two processes have the same priority, then ties will be broken by FCFS scheduling. 20
- (i) Draw a Gantt chart illustrating the execution of these processes using the Round Robin (Time Quantum = 3 ms) scheduling algorithm. If a new process arrives at the same time as the time-slice of the executing process expires, then OS puts the executing process in the ready queue, followed by the new process.
- (ii) Compute the average turnaround time if the CPU is idle for 25ms-30ms, then SRTF scheduling is used.
- Q.2(a) Discuss the concept of 'starvation' in the context of CPU scheduling. Analyze a scenario where specific processes might be unfairly delayed and how aging could prevent such a situation. 15
- Q.2(b) Consider a system running ten I/O-bound tasks and one CPU-bound task. Assume that the I/O-bound tasks issue on an I/O operation once for every millisecond of CPU computing and that each I/O operation takes 10 milliseconds to complete. Also, assume that the context-switching overhead is 0.1 millisecond and that all processes are long-running tasks. Describe the CPU utilization for a round-robin scheduler when: 12
- The time quantum is 1 millisecond
 - The time quantum is 10 milliseconds
- Q.2(c) Draw the state diagram of a process from creation to termination, including all transitions. 08
- Q.3(a) What is deadlock? What are the four essential conditions for a deadlock to occur? 10
- Q.3(b) "A cycle in the graph is a necessary but not a sufficient condition for the existence of deadlock."- Explain the statement. 10
- Q.3(c) Consider the following Allocation, Current need, and Available matrix. 15

	Allocation				Max Current Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	0	0	1	2	0	0	1	2	1	5	2	0
P ₁	1	0	0	0	1	7	5	0				
P ₂	1	3	5	4	2	3	5	6				
P ₃	0	5	2	3	0	6	4	5				
P ₄	0	1	2	4	0	5	6	4				

Now, answer the following questions using the Banker's algorithm:

- (i) What is content of the matrix 'Need'?
- (ii) Is this system in a safe state?
- (iii) If a request from process P₁ arrives for (0, 3, 2, 0), Can the request be granted immediately?
- Q.4(a) Explain the steps an OS takes for handling a page fault with appropriate figures. 08
- Q.4(b) Write short notes on: 12
- (i) Thrashing (ii) Copy on Write (iii) Belady anomaly (iv) Demand Paging

- Q.4(c) Consider the following page reference string using three frames that are initially empty: 15

7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2

How many page faults would occur for the following page replace algorithm?

- (i) FIFO Page Replacement
- (ii) LRU Page Replacement
- (iii) Optimal Page Replacement

SECTION-B

- Q.5(a) What do you mean by address binding? Explain different stages of address binding. 10

- Q.5(b) Consider the following segment table with each segment's base address and limit. 10

Segment	Base Address	Limit
0	520	500
1	350	150
2	100	200
3	1800	400
4	1050	600

Now, calculate the physical address for the following logical address:

i) 2, 50 ii) 4, 700 iii) 0, 400 iv) 1, 100 v) 3, 500

- Q.5(c) Draw the paging hardware with translation load-aside buffers (TLB) 10

- Q.5(d) Write the difference between Internal and External Fragmentation. 05

- Q.6(a) How does the Peterson algorithm guarantee mutual exclusion for two processes sharing a critical section? Explain shared variables, flags, and turn variables. Discuss how their components collaborate to ensure only one process accesses the critical section at a time. 15

- Q.6(b) "An operating system is similar to a government." – Justify the statement. 10

- Q.6(c) Consider the following code segment: 10

```
Pid ← t Pid;
Pid = fork();
if (Pid == 0) {
    fork();
    thread-create(...);
}
fork();
```

- (i) How many unique threads are created?
- (ii) How many unique processes are created?

- Q.7(a) Suppose the disk request sequence for a disk with 100 cylinders is: 24

45, 20, 90, 10, 50, 60, 80, 25, and 70

Assume that the initial position of the R/W head is on cylinder number 50. Now, find the total distance the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms.

(i) SSTF, (ii) SCAN, (iii) C-SCAN, (iv) LOOK, (v) C-LOOK, and (vi) FIFO

- Q.7(b) Explain the tree-structured and acyclic graph directory structure. Briefly discuss how the acyclic-graph directory structure can resolve the problem associated with the tree-structured directory. 11

- Q.8(a) Briefly explain the many-to-many, one-to-one, and many-to-one multithreading modules. 10

- Q.8(b) Discuss three different parameter passing methods for the system call. 08

- Q.8(c) Write down the difference between RAID 0 and RAID 1. Explain RAID 0+1 and 1+0 with an example diagram. 10

- Q.8(d) What is multiprocessing? Write down the advantages of the multi-processor system. Draw the diagram of a dual-core multi-processor design. 07

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Section-A

- Q.1(a) Create a diagram illustrating the different process states and transitions between them. Explain the significance of each state and the conditions that lead to state transitions. Use your diagram to showcase the sequence of states a process might go through. 10
- Q.1(b) Compare and contrast different IPC mechanisms, such as message passing and shared memory. Describe the advantages and disadvantages of each mechanism and discuss scenarios where one might be preferred over the other. 10
- Q.1(c) How does the Peterson algorithm guarantee mutual exclusion for two processes sharing a critical section? Explain shared variable, flags and turn variable. Discuss how these components collaborate to ensure only one process accesses the critical section at a time. Highlight challenges like busy-waiting loops and suggest how the algorithm could accommodate more than two processes. 15
- Q.2(a) Consider the following information for the 6 processes that are going to be executed in a CPU. 20

Process	Priority	Burst Time	Arrival Time
P ₁	5	10	100
P ₂	30	25	25
P ₃	10	10	105
P ₄	35	15	60
P ₅	40	20	0
P ₆	30	25	30

Now, apply the FCFS, and preemptive priority scheduling algorithms to schedule the above processes on a single CPU and find out the following evaluation metrics.

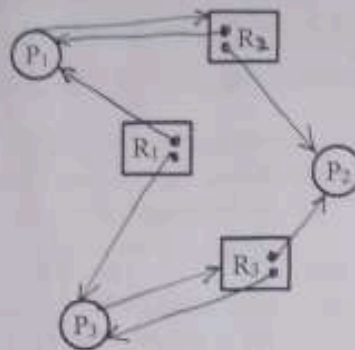
- Show the Gantt chart for each process scheduling algorithm. (Draw only the final Gantt chart for each algorithm)
 - Find the average response time, average waiting time, and average turnaround time for each algorithm.
- Q.2(b) Explain the effect of increasing and decreasing the time quantum for the round robin scheduling algorithm with a suitable example. 09
- Q.2(c) What is the difference between preemptive and non-preemptive process scheduling algorithms. 06
- Q.3(a) Explain the concept of a circular wait condition in the context of deadlock. Provide an example involving three processes and three resources to demonstrate how a circular wait condition can lead to a deadlock. 10
- Q.3(b) Consider the following snapshot of a system: 10

	Allocation				Current Need				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	0	0	1	2	0	0	1	2	1	5	2	0
P ₁	1	0	0	0	1	7	5	0				
P ₂	1	3	5	4	2	3	5	6				
P ₃	0	5	2	3	0	6	4	5				
P ₄	0	1	2	4	0	5	6	4				

Is this system in a safe state? Show your computation step-by-step and your safe sequence explicitly; Otherwise there might be a deduction in your marks.

Q.3(c) Consider the following Resource Allocation Graph (RAG):

	P_1	P_2	P_3
P_1	1	1	0
P_2	0	1	1
P_3			



- Does the above allocation graph contain a deadlock? Justify your answer.
- Assume now that P_2 also demands resource R_1 . Does this allocation graph contain a deadlock?

Q.4(a) Contrast between RAID 0 and RAID 1. How does mirroring enhance data redundancy? 05

Q.4(b) A disk drive has 3000 cylinders numbered from 0 to 2999. The current head position is at cylinder 1200, and the previous request was at cylinder 1800. The queue of pending requests, in FIFO order, is: 15

900, 1300, 600, 2400, 500

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for each of the following disk scheduling algorithms?

- FCFS
- C-SCAN
- LOOK
- SCAN
- SSTF

Q.4(c) Consider a file currently consisting of 1000 blocks. Assume that the file-control block (and the index control block, in the case of indexed allocation) is already in memory. Calculate how many disk I/O operations are required for contiguous, linked, and indexed (single-level) allocation strategies, if for one block, the following conditions hold. In the contiguous-allocation case, assume that there is no room to grow at the beginning but there is room to grow at the end. Also assume that the block information to be added is stored in memory. 15

- The block is added at the beginning.
- The block is added in the middle.
- The block is removed from the middle.
- The block is removed from the end.
- The block is added at the end.

Section-B

Q.5(a) Consider a computer system with memory partitions of sizes: 300 KB, 500 KB, 200 KB, 800 KB, and 400 KB (in order) 20

Four processes need to be allocated, each requiring memory space:

250 KB, 550 KB, 150 KB, and 600 KB (in order)

Analyze the allocation process using the First-Fit, Best-Fit, and Worst-Fit algorithms.

- Apply the First-Fit algorithm to allocate memory for the given processes. List the allocated partitions for each process.
- Apply the Best-Fit algorithm to allocate memory for the processes. For each process, identify the partition that best fits the process size and allocate it accordingly.
- Apply the Worst-Fit algorithm to allocate memory for the processes. Describe how each process would be allocated by selecting the partition that results in the largest remaining free space.
- Calculate the total amount of internal fragmentation for each algorithm.
- Determine which allocation algorithm makes the most efficient use of memory.

- Q.5(b) Briefly explain the paging hardware with TLB. Now, consider a paging system with the page table stored in memory. 15
- If a memory reference takes 50 nanoseconds, how long does a paged memory reference take?
 - If we add TLBs, and 75 percent of all page-table references are found in the TLBs, what is the effective memory reference time? (Assume that finding a page-table entry in the TLBs takes 2 nanoseconds, if the entry is present.)
- Q.6(a) Illustrate Belady's Anomaly in FIFO page replacement algorithms. [Suppose the page reference string is 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5] 08
- Q.6(b) Compare the numbers of page-faults for LRU and optimal page replacement algorithms by considering the page references as 1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6 with a 4-page frame. 12
- Q.6(c) Explain the relationship between the working set of a process and its page fault rate. 06
- Q.6(d) Briefly explain the many-to-one, one-to-one, and many-to-many multithreading models. 09
- Q.7(a) Under what circumstances do page faults occur? Describe the actions taken by the operating system when a page fault occurs. 12
- Q.7(b) Is it possible to have concurrency but not parallelism? Explain. 08
- Q.7(c) What is the cause of thrashing? Assume the working-set model is used to handle the thrashing issue and the size of the working-set window is 15. Consider the past page references of the following processes at time t. 15
- 0 $\xrightarrow{\hspace{10em}}$ T

P₀: 1, 2, 5, 2, 3, 2, 6, 5, 4, 5, 6, 2, 3, 5, 4

P₁: 2, 4, 2, 2, 1, 2, 4, 5, 1, 4, 5, 2, 3, 4, 1

P₂: 2, 2, 3, 1, 3, 2, 3, 5, 4, 5, 6, 2, 3, 1, 4

P₃: 1, 3, 1, 4, 4, 1, 4, 6, 2, 4, 5, 4, 5, 5, 2

P₄: 2, 1, 5, 3, 3, 5, 6, 1, 3, 5, 6, 2, 1, 1, 3

Fig. 7(c)
- What is the working set for each process at time T?
 - Assume we have 18 frames of physical memory. Is the system in thrashing condition?
- Q.8(a) What are the advantages of multi-processor systems? Draw a symmetric multiprocessing architecture. 10
- Q.8(b) What do you mean by the system calls? Discuss three different parameter passing methods for the system call. 10
- Q.8(c) Briefly explain the following topics – 15
- Dual-mode operation in OS
 - Types of services from cloud computing
 - Boot-strap program.

END

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SECTION-A

- Q.1(a) In a multiprogramming and time-sharing environment, several users share the system simultaneously. This situation can result in various security problems: 10
- i) What are two such problems?
 - ii) Can we ensure the same degree of security in a time-shared machine as in a dedicated machine? Explain your answer.
- Q.1(b) Discuss with examples, how the problem of maintain coherence of cached data manifests itself in the following environments: 15
- i) Single-processor systems
 - ii) Multiprocessor systems
 - iii) Distributed systems
- Q.1(c) Identify several advantages and disadvantages of open-source operating systems. List some applications appropriate for a peer-to-peer system. 10
- Q.2(a) What are the two modes of interprocess communication? What are the strengths and weaknesses of the two approaches? 10
- Q.2(b) How are iOS and Android similar? How are they different? Explain why Java programs running on Android systems do not use the standard Java API and virtual machine. 13
- Q.2(c) What are the five major activities of an OS with regard to file management? 05
- Q.2(d) Explain the concept of context switching, memory stall and soft deadline. 07
- Q.3(a) What criteria are used to measure the performance of a CPU scheduling algorithm? 10
- Q.3(b) Consider the following set of processes with the length of the CPU-burst time given in milliseconds: 16

Table: 3(b)

Process	Arrival Time	Burst Time	Priority
P_1	0	7	3
P_2	2	4	2
P_3	4	1	4 (Lowest)
P_4	5	4	1 (Highest)
P_5	3	2	1 (Highest)

- i) Draw the Gantt charts illustrating the execution of these processes using Priority Preemptive and SRTF scheduling algorithms
 - ii) Compute what the average turnaround will be if the CPU is left idle for the first 2 unit and then SRTF scheduling is used
- Q.3(c) Explain the effect of increasing the time quantum to an arbitrary large number and decreasing the time quantum to an arbitrary small number for round robin scheduling algorithm with suitable example. 09
- Q.4(a) What is a deadlock? What are the necessary conditions for a deadlock to occur? 08

Q.4(b) A system has three types of resources R_1, R_2 and R_3 and their numbers of instances are 3, 2 and 2 respectively. Four processes are currently competing for these resources in the following number: 12

- P_1 is holding one unit of R_1 and is requesting for one unit of R_2
- P_2 is holding one unit of R_2 and is requesting for one unit of each of R_1 and R_3
- P_3 is holding one unit of R_1 and is requesting for one unit of R_2
- P_4 is holding two units of R_3 and is requesting for one unit of R_1

Determine which if any of these processes are deadlock in this state.

Q.4(c) Consider the following snapshot of a system: 15

	Allocation				Max			
	A	B	C	D	A	B	C	D
P_0	3	0	1	4	5	1	1	7
P_1	2	2	1	0	3	2	1	1
P_2	3	1	2	1	3	3	2	1
P_3	0	5	1	0	4	6	1	2
P_4	4	2	1	2	6	3	2	5

Fig. 4(c)

Using the banker's algorithm, determine whether each of the following states is unsafe. If the state is safe, illustrate the order in which the processes may complete. Otherwise, illustrate why the state is unsafe.

- Available = (0,3,0,1) ✗
- Available = (1,0,0,2) ✓

SECTION-B

- Q.5(a) Why page sizes are always a power of 2? 05
- Q.5(b) With a neat sketch, explain how logical address is translated into physical address using paging mechanism. 08
- Q.5(c) Given memory partitions of 500K, 100K, 200K, 300K and 600K (in order), how would each of the First-fit, Best-fit and Worst-fit algorithms place processes of size 418K, 202K, 505K, 112K and 96K (in order)? Which algorithm makes the most efficient use of memory? 14
- Q.5(d) Consider the following segment table shown in Table 5(d). What are the physical addresses for the following logical addresses? 08

Table: 5(d)

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	95

0,430
2,500
3,680
4,90

- Q.6(a) What happens if a process tries to access a page that was not brought in memory? Explain the steps in handling a page fault with relevant figure. 15

Q.6(b) Consider the following page reference string:

1,2,3,4,5,1,2,3,4,1,2,3,5,1,2,3

How many page faults would occur for the following page replacement algorithms assuming four frames?

- i) LRU page replacement
- ii) Optimal page replacement

08

Q.6(c) What is an inverted page table? Briefly explain the structure of hashed page table.

10

Q.7(a) Consider a paging system with page table stored in memory:

- i) If a memory reference takes a 50ns how long does a page memory reference take?
- ii) If we add TIB and 75% of all page table reference are found in TLB, what is the effective memory reference time? (Assume that finding a page entry in TLB TAKE 0.000002 MS, if entry is present)

09

Q.7(b) Compare RAID0, RAID1 and RAID5 in terms of performance improvement, ability to withstand failure and space overhead required to implement the particular RAID level.

16

Q.7(c) On a disk with 200 cylinders, numbered 0 to 199, compute the number of tracks the disk arm must move to satisfy the entire request in the disk queue. Assume the last request received at track 100. The queue in FIFO order contains requests for the following tracks 55, 57, 37, 15, 80, 160, 150, 36, 185. Perform the computation to find the seek time for the following disk scheduling algorithms:

- i) FCFS
- ii) SSTF
- iii) C-SCAN
- iv) LOOK

Q.8(a) Consider the following code segment:

10

```
pid_t pid;
pid = fork ();
if (pid == 0) \*Child process*\
{
    fork ();
    thread_create (.....);
}
fork();
```

- i) How many unique processes are created?
- ii) How many unique threads are created?

Q.8(b) Is it possible to have concurrency but not parallelism? Explain.

10

Q.8(c) Explain the differences in how much the following scheduling algorithm discriminate in favor of short processes:

15

- i) FCFS
- ii) RR
- iii) Multilevel feedback queues